

PETSc vs MATLAB 3D SSR Performance Comparison

slope_stability

Indirect SSR, P2 elements, MPI-8 PETSc vs OMP-8 MATLAB

Repository:	slope_stability checkout
Study root:	docs/petsc_matlab_performance_comparison
Performance artifacts:	artifacts/petsc_matlab_performance_comparison
Seepage artifacts:	artifacts/benchmarks/mpi8

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1 Methodology

The study definition uses indirect SSR only and P2 elements only. It covers homogeneous 3D SSR, heterogeneous 3D SSR, and the benchmarked heterogeneous 3D seepage SSR COMSOL-transition case.

PETSc runs use `mpirun-n8` with `OMP_NUM_THREADS=1`. MATLAB runs use `OMP_NUM_THREADS=8` and the BoomerAMG MATLAB path with 8 threads. The dry PETSc and MATLAB runs for a given case share the same continuation horizon, estimated by a PETSc smoke run on the coarsest physical mesh with `d_lambda_diff_scaled_min=0.01`. Main runs then disable the continuation delta-lambda stop and terminate on $\omega_{\max}$. The seepage section uses the existing benchmark-suite COMSOL-transition SSR artifacts with `lambda_init=1.0`, `d_lambda_init=0.1`, and `d_lambda_diff_scaled_min=0.005`.

The heterogeneous 3D case adds one appendix variant where PETSc Newton stopping changes from relative residual to absolute delta-lambda with tolerance 10^{-4} .

The homogeneous and heterogeneous dry cases use the requested relative-residual Newton tolerance 10^{-4} . The normalized seepage inputs use the benchmarked COMSOL-transition PETSc/MATLAB pair and exclude the waterlevels/concave study rows until that separate seepage parity issue is resolved.

2 Environment And Outputs

The report consumes only normalized CSV files under `data/`. Figures are generated as vector PDFs under `figures/`, and table fragments are generated under `generated/`. Long-running raw solver outputs remain under `artifacts/petsc_matlab_performance_comparison`. Figures and tables include only completed PETSc/MATLAB runs captured in those CSV files.

3 Homogeneous 3D SSR

Homogeneous 3D SSR

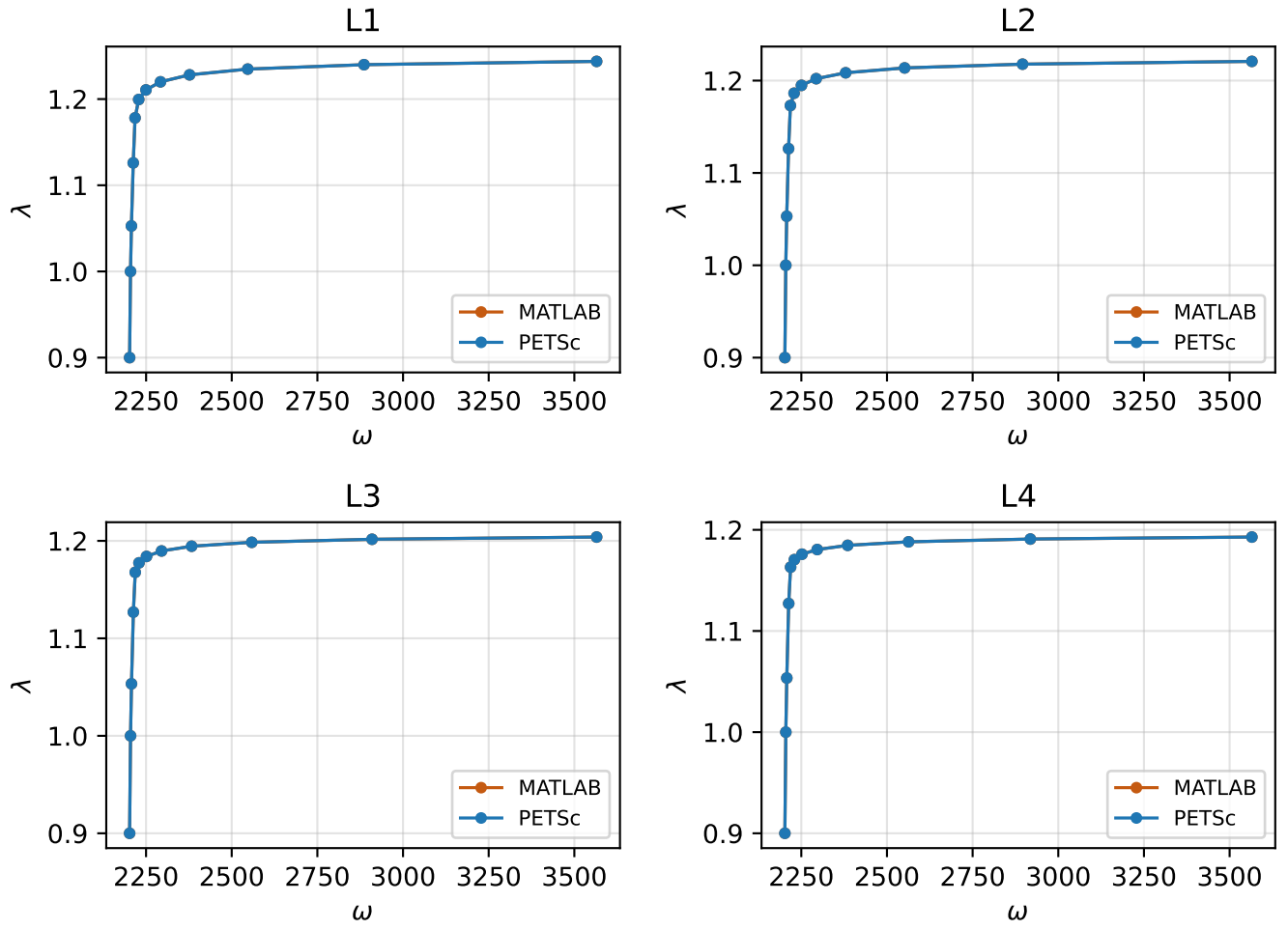


Figure 1: Continuation curves by mesh level for the homogeneous 3D SSR case.

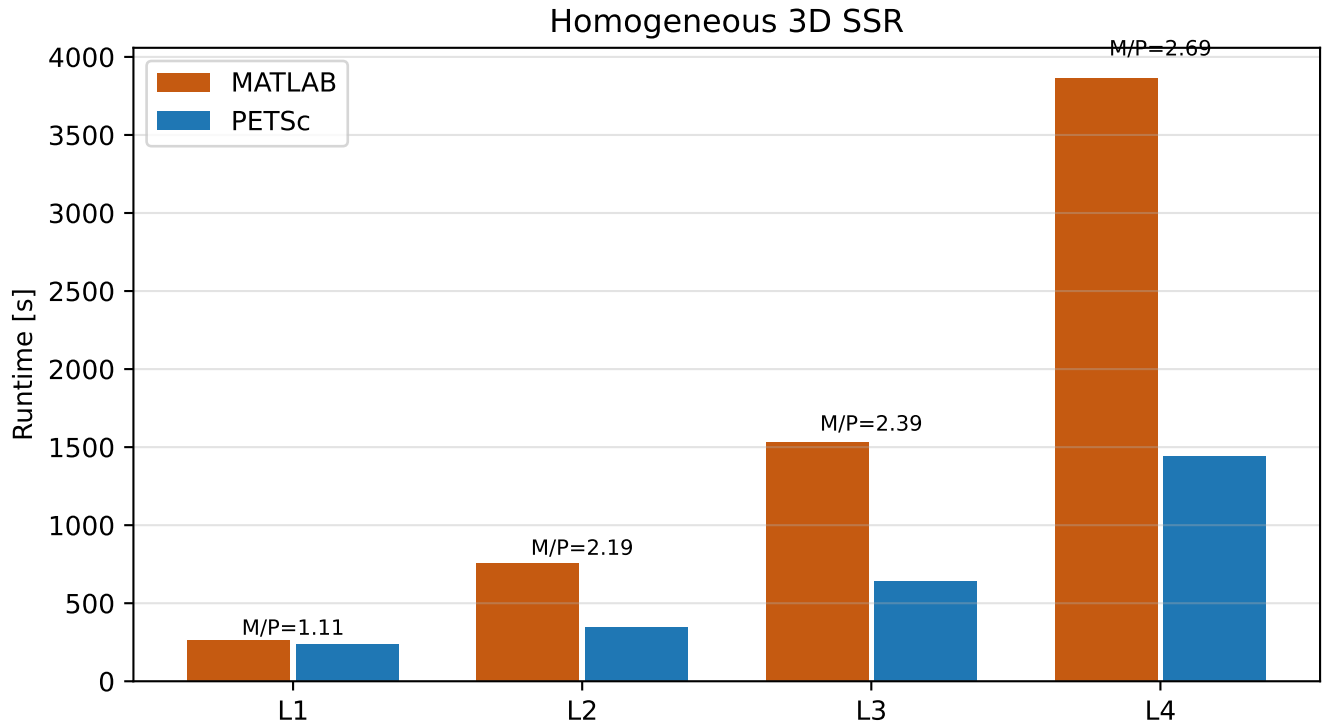


Figure 2: Runtime comparison by mesh level for the homogeneous 3D SSR case.

Table 1: Homogeneous 3D SSR summary.

Level	Unknowns	PETSc [s]	MATLAB [s]	M/P	PETSc steps	MATLAB steps	Stop (P/M)
L1	71947	234.857	261.782	1.11	12	12	omega max / omega max
L2	138382	345.950	759.187	2.19	12	12	omega max / omega max
L3	264607	640.670	1531.512	2.39	12	12	omega max / omega max
L4	508962	1438.968	3865.076	2.69	12	12	omega max / omega max

4 Heterogeneous 3D SSR

Heterogeneous 3D SSR

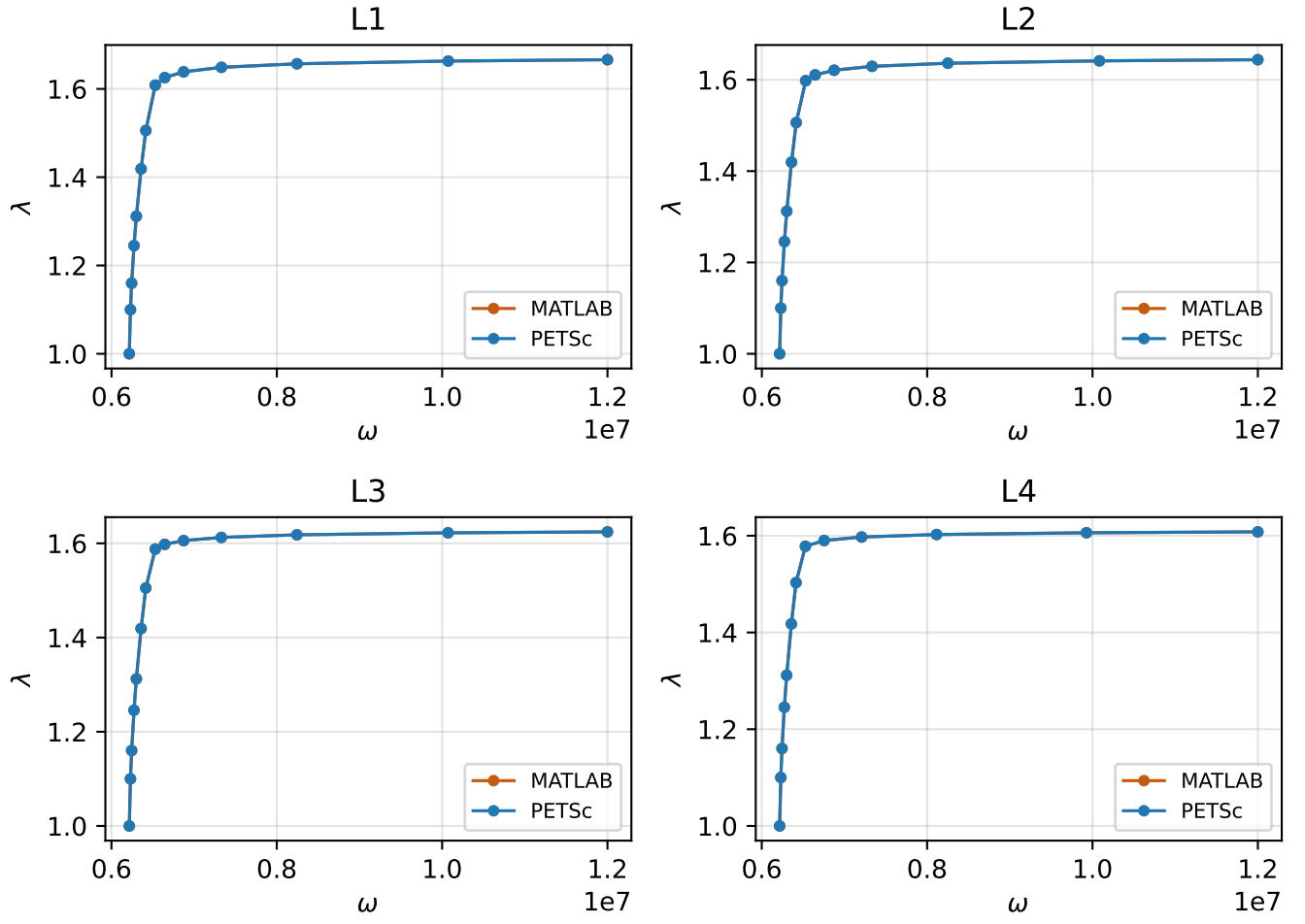


Figure 3: Continuation curves by mesh level for the heterogeneous 3D SSR case.

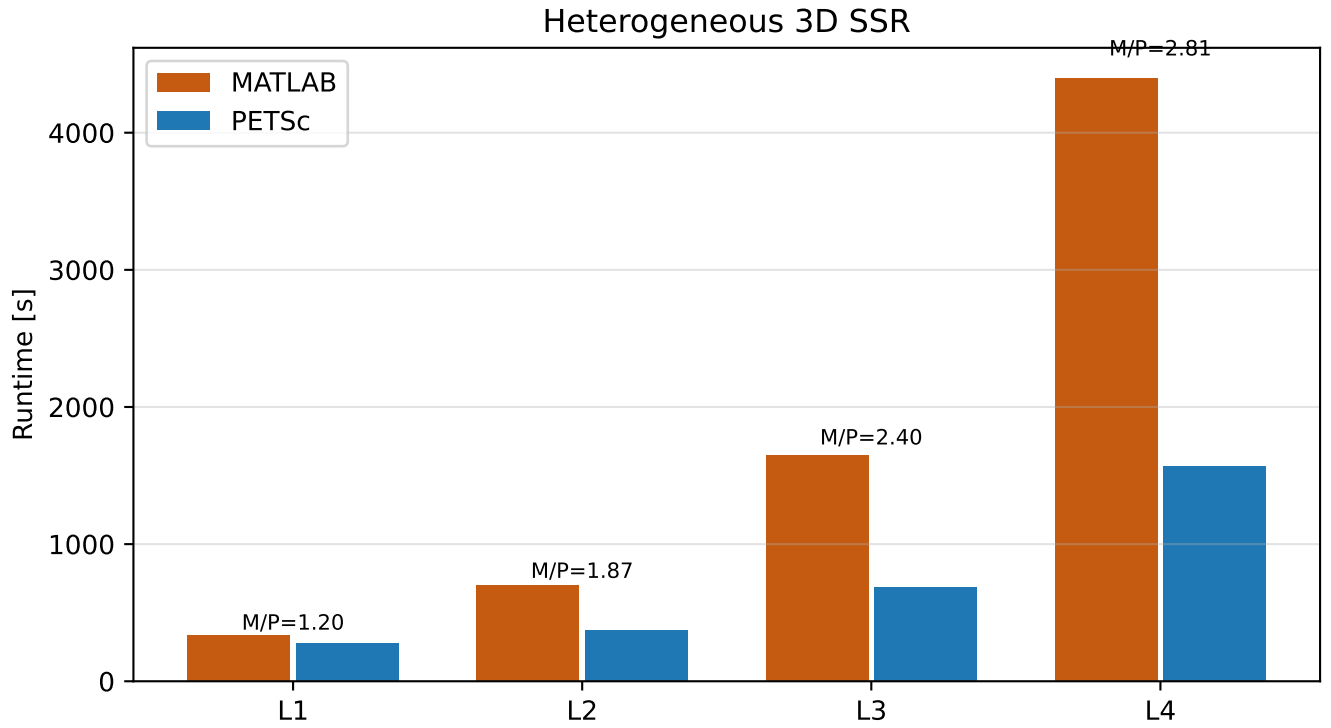


Figure 4: Runtime comparison by mesh level for the heterogeneous 3D SSR case.

Table 2: Heterogeneous 3D SSR summary.

Level	Unknowns	PETSc [s]	MATLAB [s]	M/P	PETSc steps	MATLAB steps	Stop (P/M)
L1	80362	279.519	334.248	1.20	14	14	omega max / omega max
L2	145298	375.089	700.772	1.87	14	14	omega max / omega max
L3	265739	685.231	1645.273	2.40	14	14	omega max / omega max
L4	490015	1564.466	4399.089	2.81	13	13	omega max / omega max

5 Heterogeneous 3D Seepage SSR

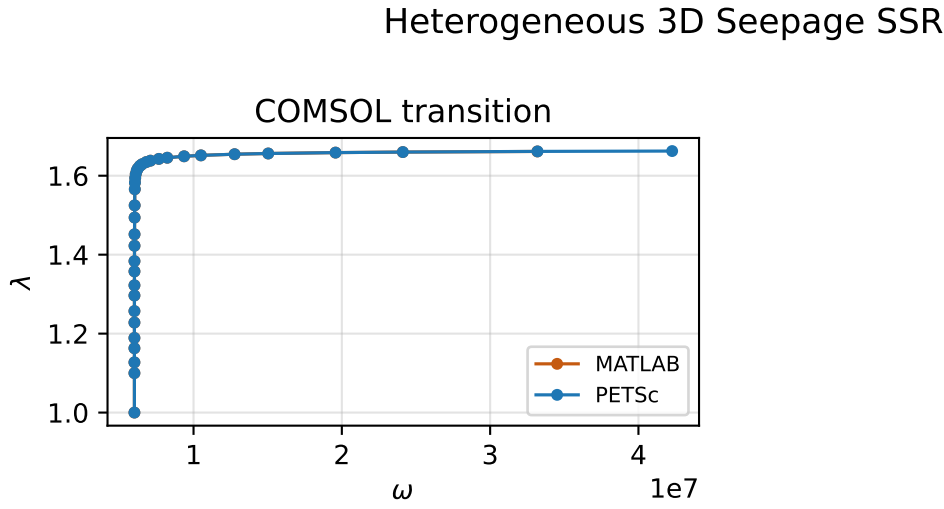


Figure 5: Continuation curves by mesh level for the heterogeneous 3D seepage SSR case.

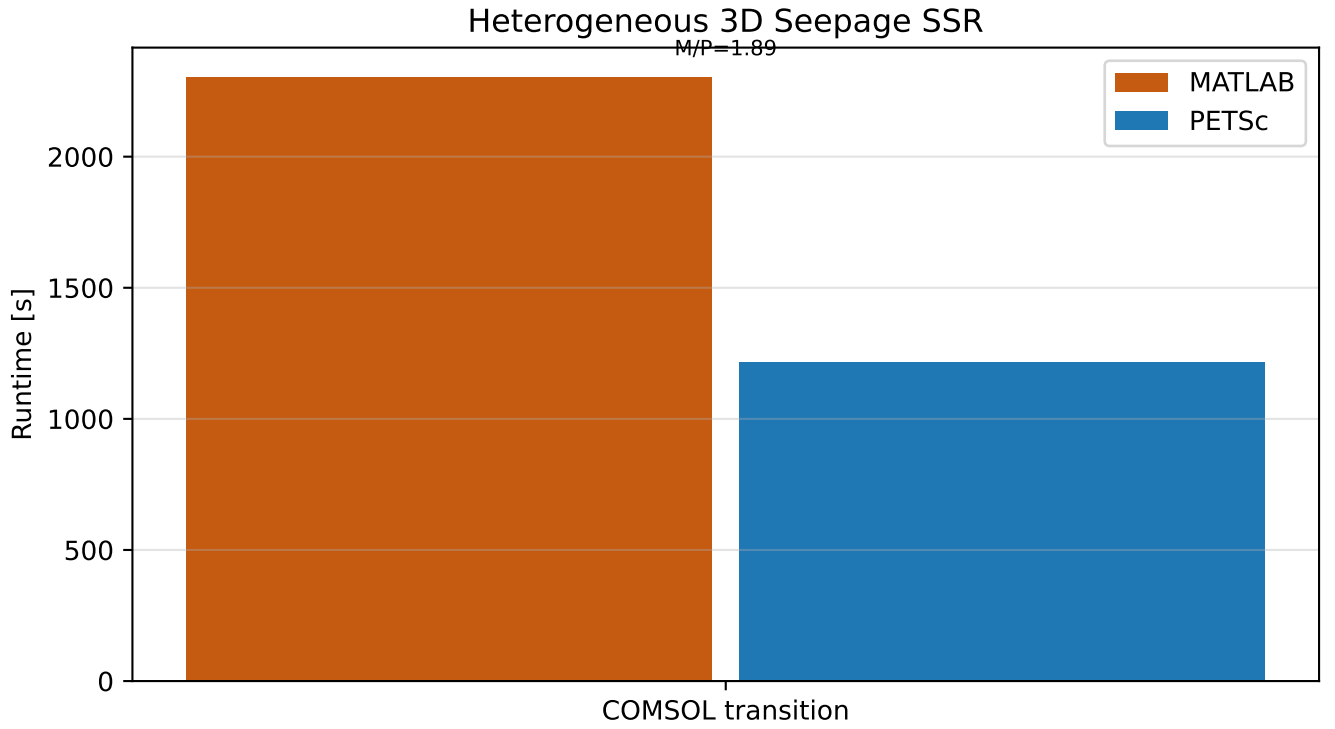


Figure 6: Runtime comparison by mesh level for the heterogeneous 3D seepage SSR case.

Table 3: Heterogeneous 3D Seepage SSR summary.

Level	Unknowns	PETSc [s]	MATLAB [s]	M/P	PETSc steps	MATLAB steps	Stop (P/M)
COMSOL transition	231201	1217.168	2300.903	1.89	35	34	$\Delta\lambda_{\min}$ / $\Delta\lambda_{\min}$

The seepage section reports the benchmark-suite COMSOL-transition pair from `run_3D_hetero_seepage_SSR_comsol_capture`. This is the current PETSc/MATLAB SSR seepage parity surface: both histories start from the configured continuation seed path at $\lambda = 1.0$ and follow the same accepted continuation curve. The experimental waterlevels/concave rows are excluded from this report because their seepage field parity is not yet verified.

A Heterogeneous 3D Newton Stopping Appendix

Heterogeneous 3D SSR Appendix

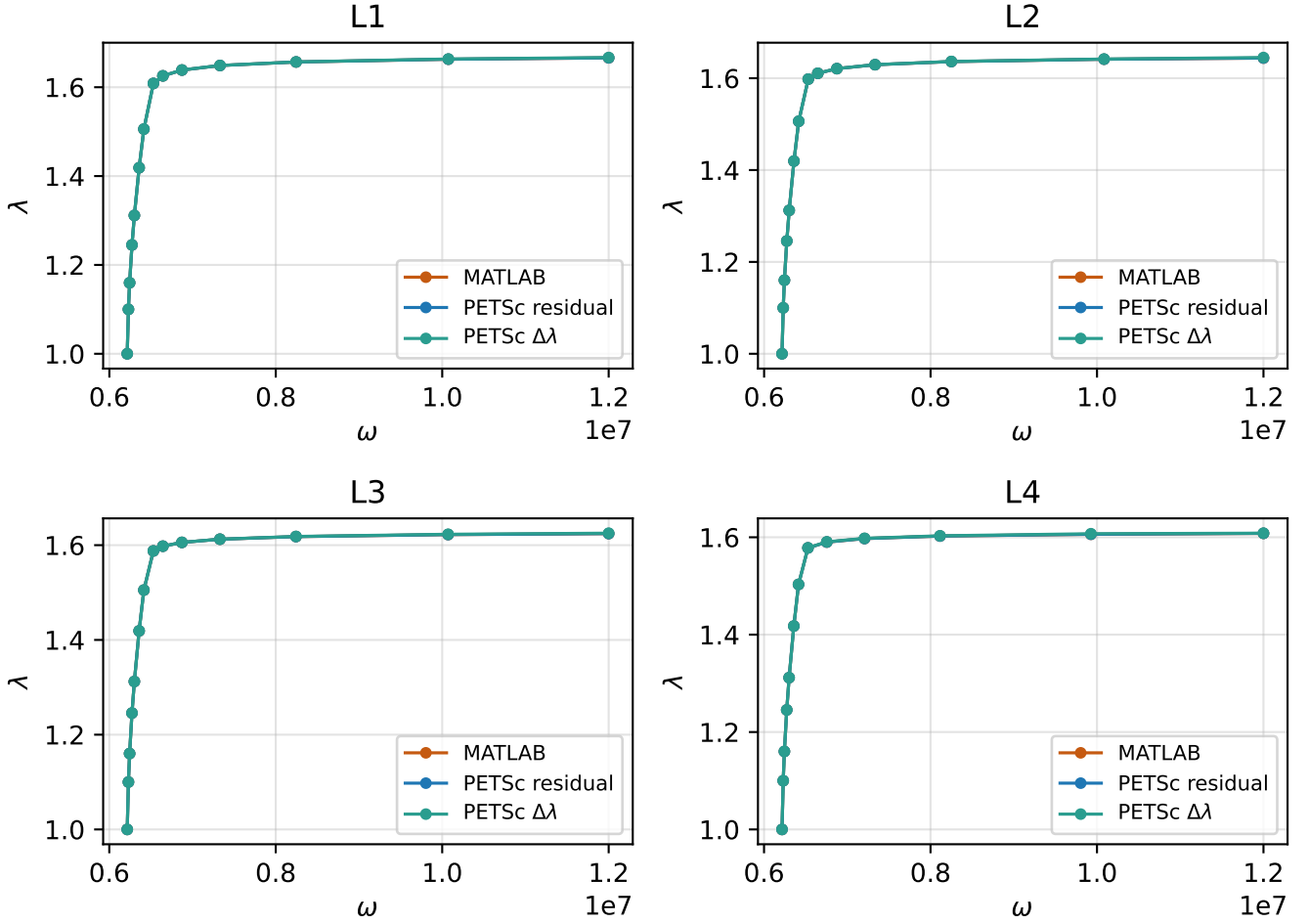


Figure 7: Continuation curves for heterogeneous 3D SSR with the PETSc delta-lambda appendix variant.

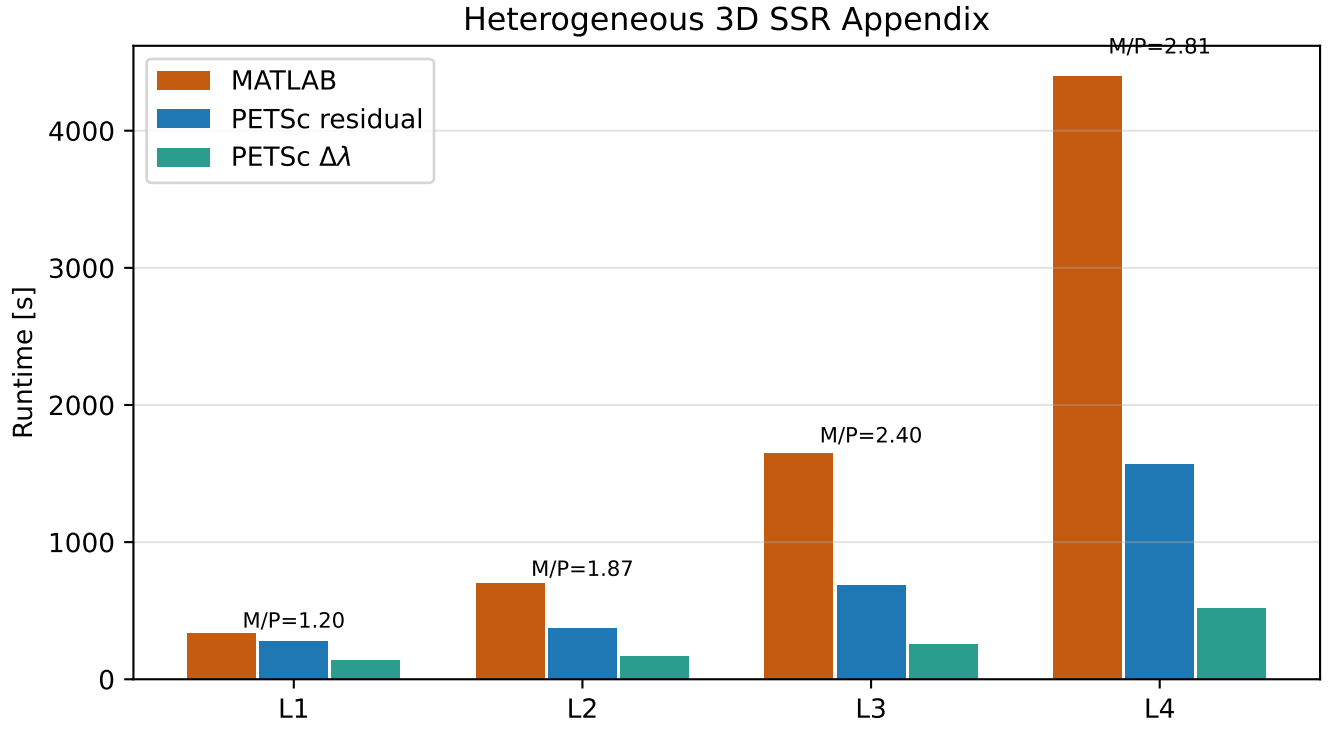


Figure 8: Timing comparison for heterogeneous 3D SSR with the PETSc delta-lambda appendix variant.

Table 4: Heterogeneous 3D SSR appendix: PETSc residual-stop vs PETSc delta-lambda-stop.

Level	PETSc res. [s]	PETSc $\Delta\lambda$ [s]	MATLAB [s]	Res. steps	$\Delta\lambda$ steps	MATLAB steps
L1	279.519	136.049	334.248	14	14	14
L2	375.089	167.123	700.772	14	14	14
L3	685.231	257.089	1645.273	14	14	14
L4	1564.466	520.176	4399.089	13	13	13